Hermes OSINT Hybrid Framework

Version 1.3 - Modular evidence-first OSINT for domains, companies, public identities, usernames, and vendor risk

Design goal

Keep the strong evidence/provenance engine from the domain workflow, while adding modular due-diligence and fuzzy-search capabilities without turning Hermes into one oversized prompt. The master skill routes each case to the smallest relevant module.

Important: This framework uses public-source and passive methods by default. References to FBI/DOJ describe correlation patterns visible in public materials, not an official internal FBI manual.

Module	Primary purpose
domain	DNS, RDAP, CT, infrastructure, archives, technical pivots
company	Legal entity, filings, ownership signals, public reputation
person	Public professional identity verification with privacy minimization
username	Public handle discovery plus corroboration
vendor	Third-party cyber/privacy due diligence and evidence requests

1. Architecture

The framework is a router plus specialized modules. Load only what the case requires.

```
osint-hybrid ■■■ domain ■■■ company ■■■ person ■■■ username ■■■ vendor
```

• Master router classifies the seed as exact, partial, ambiguous, or single-pivot.

• Fuzzy search is controlled and produces candidates, not facts.

• Validated artifacts move between modules through an explicit handoff contract.

• All modules use the same evidence schema and confidence vocabulary.

• The domain module retains the runnable passive collector from v1.2 and adds signal strength plus Mermaid graph output.

Why this matters for local models

A modular design reduces context pressure. A domain investigation does not need the entire company/person/vendor playbook in context, and a company filing check does not need every DNS or certificate rule.

2. Evidence model

Every material observation should preserve provenance so another analyst can reproduce the path from source to finding.

Field	Meaning
evidence_id	Stable case-local identifier
source_adapter / source_name	Where the observation came from
query	Exact lookup or retrieval seed
collected_utc	When the analyst/tool collected it
observed_utc	Historical time represented by the source, if known
artifact_type / artifact	Domain, IP, certificate, company, username, etc.
relationship	What was directly observed
raw_ref	Path/reference to preserved raw material
signal_strength	Weak / moderate / strong distinctiveness
confidence	Confidence in the observation itself

Analytical separation

• OBSERVATION: directly supported by the source.

• INFERENCE: analyst interpretation built from evidence.

• UNKNOWN: a question the available evidence does not answer.

• ALTERNATIVE_EXPLANATION: plausible reason the observed relationship may not mean common control/identity.

> One indicator is never enough for attribution.

3. Confidence and false-positive controls

Signal	Default strength	Reason
Same CDN/shared IP	Weak	Co-tenancy is common and often unrelated.
Same generic DNS provider	Weak	Large providers host unrelated customers.
Same uncommon/custom nameservers	Moderate	More distinctive, especially with historical continuity.
Same custom MX	Moderate	Can indicate shared administration but may be managed service.
Same certificate context	Moderate	Useful technical relation; still not ownership by itself.
Same unique public contact identifier across history	Strong	Distinctive when independently verified.
Same authoritative legal entity	Strong	Prefer registry/filing evidence.
Multi-source historical continuity	Strong	Migration/rebrand continuity can materially strengthen a relationship.

Relationship confidence

Label	Use
CONFIRMED	2+ independent high-quality sources, no material contradiction.
PROBABLE	One strong source plus corroboration, or multiple sources with minor inconsistency.
POSSIBLE	Plausible but limited corroboration; alternatives remain credible.
UNVERIFIED	Lead only.
DISCONFIRMED	Stronger evidence contradicts the candidate relationship.

4. Fuzzy search without identity drift

Fuzzy search is useful only when tightly bounded. Generated variants are search candidates, not facts.

Input state	Action
EXACT_VERIFIED	Direct query; avoid unnecessary expansion.
PARTIAL	Generate limited variants; verify against authoritative/public context.
AMBIGUOUS	Widen once, cluster 5-15 candidates, then narrow with known attributes.
SINGLE_PIVOT	Move only to logically adjacent artifacts.

Stop rules

- Stop widening near 5-15 candidates.
- Stop narrowing when 1-3 candidates remain and each has at least two corroborating attributes.
- Do not merge company/person records on name similarity alone.
- Preserve rejected candidates when they explain why a tempting match was discarded.

5. Domain module

The domain module is the technical core. It remains passive/public by default and creates a structured case directory.

```
python3 bin/domain_osint.py example.com
```

Collectors

Adapter	Purpose
RDAP	Authoritative registration/status metadata via service discovery
DNS / dig	A, AAAA, NS, MX, TXT, CNAME, SOA, CAA
Certificate Transparency	Public certificate/SAN history
Wayback	Historical public web captures
urlscan.io	Historical/public page-scan context
Hunt.io	Optional third-party domain enrichment
Shodan	Optional passive host enrichment for already-observed IPs
Sherlock	Optional public username discovery for analyst-supplied handles

Outputs

```
case.yaml evidence.jsonl findings.yaml raw/ screenshots/ graph.mmd report.md
```

6. Company module

• Seed validation: legal name, jurisdiction, registration number, brand/domain aliases.

• Legal identity from authoritative company registries/filings first.

• Public ownership/officer signals, financial/public filings, relevant regulatory/legal records.

• Reputation and media with allegation/fact separation.

• Validated domains hand off to the domain module.

• Network correlation only after entity-resolution checks.

Entity resolution

Prefer exact legal identifiers or multiple matching attributes. Legal name + jurisdiction, registration number, or legal name + officer + address is substantially better than simple string similarity.

> Do not turn a brand match into a legal-entity conclusion without registry/filing support.

7. Public identity and username modules

The person module is intentionally narrower than a generic "background check." Its purpose is lawful public/professional identity verification, with privacy minimization.

Person workflow

- Define a legitimate verification question.
- Use only seed attributes needed for disambiguation.
- Prefer authoritative/professional sources.
- Corroborate common names with at least two public attributes.
- Omit unnecessary private/sensitive information from the report.

Username workflow

- Treat a matching handle as a candidate link, not proof of identity.
- Use Sherlock/Maigret-style discovery only as leads.
- Verify the public source page when possible.
- Corroborate with public bio/project/employer/cross-link context.
- Reduce confidence for short/common handles and placeholder accounts.

8. Vendor risk module

Vendor risk is evidence collection and question generation, not a compliance certification.

Phase	What to establish
Identity	Correct legal/vendor entity and validated domains
Privacy	Public privacy policy, DPA/subprocessor/data-residency claims
Assurance	Vendor claims vs independently verifiable certificate/register evidence
Incident history	Reliable public reports and regulator notices
Technical hygiene	Passive external observations via the domain module
Resilience	Public status/SLA/BCP/DR claims where available
Contract follow-up	Questions requiring vendor or legal confirmation

> Do not label a vendor "safe" or "unsafe" from OSINT alone; report factual issues, severity, uncertainty, and evidence gaps.

9. Cross-module handoff

Only validated or clearly marked speculative artifacts should move between modules.

```
artifact: type: domain|company|username|email|ip|certificate|person value: example.com source_evidence_ids:
[CASE-...-E0001] reason_for_pivot: "Observed in authoritative/public source" confidence: probable
restrictions: - passive-public
```

This handoff contract prevents a weak candidate from silently becoming a trusted seed in the next module.

10. Reporting

• EXECUTIVE: concise findings, uncertainty, and next lawful steps.

• ANALYST: evidence table, timeline, pivots, contradictions, alternative explanations.

• TECHNICAL: raw relationships, infrastructure details, provenance, tool status.

A good finding

```
finding: "Two domains appear technically related" relationship_confidence: probable supporting_evidence:
[E0004, E0012, E0021] contradictory_evidence: [] alternative_explanations: - "Could be the same
managed-service provider" unknowns: - "Common ownership not independently established"
```

The Mermaid graph (`graph.mmd`) is an analyst aid. Graph edges should not be interpreted as ownership unless the evidence explicitly supports ownership.

11. Installation and test plan

Install the package into the Hermes research skills directory:

```
bash bin/install-hermes.sh
```

First controlled domain test:

```
python3 bin/domain_osint.py
```

Review in this order

• case.yaml - which adapters actually ran and which were unavailable/rate-limited.

• evidence.jsonl - whether observations are normalized correctly.

• raw/ - verify the normalized records against source responses.

• graph.mmd - inspect whether edges are meaningful or merely shared infrastructure.

• findings.yaml - confirm automated candidate relationships are not overstated.

• report.md - check observation/inference/unknown separation.

The first real test should focus on false positives and confidence calibration, not on maximizing the number of discovered artifacts.

12. Sources and attribution

• Berkeley Protocol on Digital Open Source Investigations - OHCHR / UC Berkeley Human Rights Center: https://humanrights.berkeley.edu/publications/berkeley-protocol-on-digital-open-source-investigations/

• NousResearch Hermes Agent official OSINT Investigation skill: https://github.com/NousResearch/hermes-agent/tree/main/optional-skills/research/osint-investigation

• Hermes skill documentation / repository: https://github.com/NousResearch/hermes-agent/

• mtjikuzu/hermes-osint-skill (MIT): https://github.com/mtjikuzu/hermes-osint-skill

• Public FBI/DOJ cyber alerts, affidavits, advisories and press releases may be used as examples of public infrastructure/identifier correlation. This framework does not claim access to or reproduce an internal FBI domain-investigation manual.

Licensing note

v1.3 takes architectural inspiration from the MIT-licensed mtjikuzu project rather than copying its entire skill. The package includes ATTRIBUTION.md so future redistribution can preserve provenance and licensing context.